

VISHESH KUMAR

PhD Candidate in Data Science and Engineering | Trustworthy Machine Learning and Computer Vision

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Research Profile

PhD candidate at IISER Bhopal working at the intersection of trustworthy machine learning, representation learning, computer vision, adversarial robustness, and explainable AI. My research studies how learned representations fail under natural corruptions and adversarial attacks, and develops robust, interpretable, and generalizable learning and evaluation methods. I have published at ICML, ECCV, CVPR, AISTATS, IJCB, AAAI, and in IEEE Transactions on Big Data, and completed a six-month visiting scholarship at Rice University. I am particularly interested in robust and efficient foundation models, scalable evaluation, multimodal learning, and responsible AI.

Education

Indian Institute of Science Education and Research (IISER) Bhopal

PhD in Data Science and Engineering

Supervisor: Dr. Akshay Agarwal

12/22 – 12/26 (*expected*)

Expected graduation: 12/26

Indian Institute of Engineering Science and Technology (IIST) Shibpur

M.Sc. in Applied Mathematics

07/19 – 08/21

CGPA: **8.8/10**

Motilal Nehru College, University of Delhi

B.Sc. in Physical Science

07/16 – 06/19

CGPA: **7.71/10**

Research Experience

Visiting Scholar, Rice University, Houston, USA

01/26 – 06/26

- Conducted research with Dr. Guha Balakrishnan on interpretable and robust visual learning, focusing on unified adversarial-patch detection and localization.
- Developed prototype-based and attention-based mechanisms for identifying adversarial regions while preserving interpretable evidence for model decisions.
- Collaborated across institutions on research design, large-scale experimentation, technical writing, and publication-oriented evaluation.

PhD Researcher, Department of Data Science and Engineering, IISER Bhopal

12/22 – Present

- Formulate and execute research in adversarial robustness, corruption robustness, representation analysis, explainable AI, and trustworthy computer vision under the supervision of Dr. Akshay Agarwal.
- Design reproducible evaluation pipelines for deep neural networks across architectures, datasets, attack families, natural corruptions, and unseen distribution shifts.
- Develop methods for robust representation learning, generalized adversarial-patch detection, interpretable localization, and automated model auditing.
- Publish and present research at leading machine learning and computer vision venues, including ICML, ECCV, CVPR, IEEE TBD, and AISTATS.

Junior Research Fellow, DA-IICT, Gandhinagar, India

08/22 – 12/22

- Worked with Prof. Nabin Kumar Sahu on the SERB-funded project “Dynamical Sampling and Representation of Frames via Iterated Operator System.”
- Studied operator-theoretic and functional-analytic foundations of dynamical representations, contributing to a publication in *La Matematica*.

Selected Research Projects

Feature Collapse Under Corruption: Entropy-Based Robust Representation Learning

ICML 2026

- Investigated why deep models fail under natural corruptions through layer-wise entropy analysis, identifying feature collapse in deeper representations.
- Developed Dem-HEC, a training framework using high-entropy sample generation, contrastive learning, and knowledge distillation to improve corruption robustness across convolutional and transformer architectures.
- Built large-scale experimental pipelines spanning CIFAR, Tiny-ImageNet, and ImageNet subsets, with systematic ablations and architecture-level analysis.
- Code: github.com/visheshrajput2408/Dem-HEC.

Breaking Rigidity in Adversarial Patch Attacks

ECCV 2026

- Introduced deformable universal adversarial patches that move beyond fixed rectangular geometry through smooth spline-based shape optimization.

- Evaluated transferability and attack effectiveness across CNNs, vision transformers, object detectors, vision-language models, and segmentation systems.
- Released a reproducible benchmark and codebase for studying realistic, non-rigid physical adversarial threats.
- Code and dataset access: github.com/visheshrajput2408/SmuDPatch.

Generalized and Interpretable Defense Against Adversarial Patches *CVPR 2025; AISTATS 2026; ongoing*

- Developed AdvPatchXAI and subsequent generalized defenses for detecting unseen adversarial patches under clean and corruption-augmented conditions.
- Designed interpretable localization methods that identify adversarial regions while providing evidence for detection decisions.
- Extended this research toward PIPShield, a prototype-based unified detection-localization framework with clean and anomaly prototype banks.
- AdvPatchXAI code: github.com/visheshrajput2408/AdvPatchXAI.

Agentic AI System for Automated Robustness Auditing

2026 – Present

- Building an agentic evaluation platform that accepts a dataset and model checkpoint, selects relevant stress tests, executes attacks and corruptions, and produces a structured vulnerability report.
- The system is designed to summarize accuracy degradation, attack success, failure examples, and interpretable diagnostics for research and deployment settings.

Technical Skills

Programming and ML: Python, PyTorch, NumPy, pandas, scikit-learn, OpenCV, Jupyter

Research Methods: Deep learning, representation analysis, adversarial attacks and defenses, contrastive learning, knowledge distillation, optimization, model evaluation, statistical analysis

Tools: Git/GitHub, Linux, LaTeX, experiment tracking, reproducible research workflows

Selected Publications

1. **V. Kumar**, G. Balakrishnan, and A. Agarwal. “Breaking Rigidity in Adversarial Patch Attacks.” *European Conference on Computer Vision (ECCV)*, 2026.
2. **V. Kumar** and A. Agarwal. “Feature Collapse Under Corruption: An Entropy Perspective on Robust Neural Networks.” *International Conference on Machine Learning (ICML)*, 2026.
3. **V. Kumar** and A. Agarwal. “The Unseen Adversaries: Robust and Generalized Defense Against Adversarial Patches.” *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.
4. **V. Kumar**, S. Shukla, and A. Agarwal. “Robustness Benchmarking of Convolutional and Transformer Architectures for Image Classification.” *IEEE Transactions on Big Data*, 2025. [\[paper\]](#)
5. **V. Kumar** and A. Agarwal. “A Unified, Resilient, and Explainable Adversarial Patch Detector.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. [\[paper\]](#)
6. **V. Kumar**, A. Chanda, P. Bhattacharya, and A. Agarwal. “Q-MoFusion: A Quantum Classifier for Mosquito Species Classification.” *AAAI Conference on Artificial Intelligence, Student Abstract Track*, 2026.
7. N. K. Sahu and **V. Rajput**. “Dynamical Representation of Frames in Tensor Product of Hardy Spaces.” *La Matematica*, 2026. [\[preprint\]](#)
8. **V. Kumar** and A. Agarwal. “Are Object Recognition Models Effective and Unbiased for Biometric Recognition?” *IEEE International Joint Conference on Biometrics (IJCB)*, 2024.
9. **V. Kumar** and A. Agarwal. “On Unconstrained Ear Recognition for Privacy-Preserving Authentication.” *WAMWB, ACM MobileHCI Workshop*, 2023.

Manuscript Under Review

- **V. Kumar**, G. Balakrishnan, and A. Agarwal. “PIPShield: A Unified Interpretable Prototype-Based Defense for Adversarial Patch Detection and Localization.” Under review, 2026.

Awards and Fellowships

- ICML 2026 Financial Aid Award; ACM/IARCS Travel Grant for AISTATS 2026.
- Mehta Rice Engineering Scholars Program, Rice University, 2025–2026.
- CVPR 2025 DEI Travel Grant; ACM/IARCS Travel Grant for CVPR 2025; ConfAI 2025 Travel Grant.
- Visvesvaraya PhD Fellowship, Government of India.
- Qualified IIT JAM (2019, 2020) and GATE (2022).

References

Dr. Akshay Agarwal, Assistant Professor, IISER Bhopal; PhD Supervisor. akagarwal@iiserb.ac.in

Dr. Guha Balakrishnan, Assistant Professor, Rice University; Visiting-Scholar Host and Research Collaborator. guha@rice.edu